ahmad_mahmoud.resume.ts
SENIOR SOFTWARE ENGINEER ● 12+ YRS

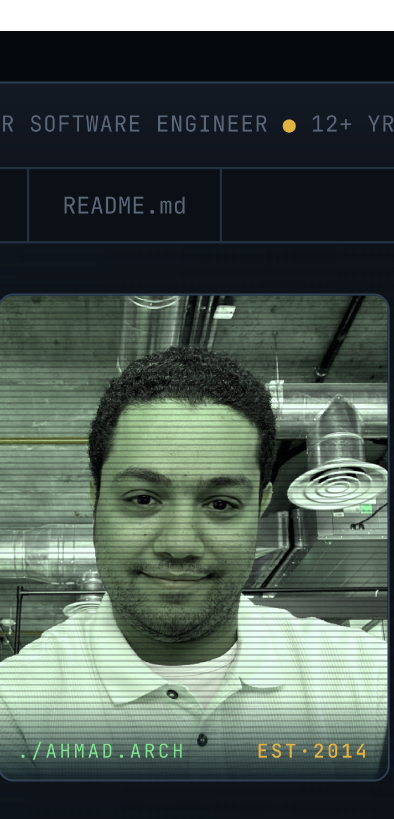
● senior_engineer.ts | experience.json | projects/ | stack.config | README.md

→ whoami --full

Ahmad Abdul-Jawad Mahmoud

Senior Software Engineer / Founder & CTO, Buraq

// **12+ years** shipping software · **iOS** · **Android** · **backend** · Cairo ⟶ Riyadh · apps used by 35M+ people



12+	35M+	99.8%+	10k+
YEARS HANDS-ON	USERS REACHED	CRASH-FREE IN PROD	TESTS ACROSS BURAQ

```
/**
* Senior Software Engineer — 12+ years shipping production
* software, native iOS first and now across Android, backend and web.
* Apps used by 35M+ people: HungerStation, Ego, and seven years on the
* Marn POS (20k+ restaurants) — an offline-first CoreData core with cloud
* sync at a 99.8–99.9% crash-free rate.
* Since March 2026: Founder & CTO of Buraq (buraqegypt.com), a ride-hailing platform
* built end to end, solo — NestJS / PostgreSQL / Redis, a Next.js console, web, iOS & Android.
*/
```

## how_i_engineer()

4 pillars

[01] systems design

Design for the failure case

An offline-first CoreData + cloud-sync core so a venue keeps trading when the network does not. A **trip state machine**, **scored dispatch** over a Redis geo index and a **double-entry ledger** with idempotent mutations, so a retry can never double-charge.

◆ Offline-first · idempotent · double-entry

[02] quality you can measure

Tests gate the merge

Automated tests that block bad merges, CI/CD built from nothing (**60% faster deployments**), and a **99.8–99.9% crash-free rate** in production. At Buraq: **2,900+ backend**, **1,000+ iOS** and **4,000+ Android** tests.

◆ 99.8–99.9% crash-free · 60% faster deploys

[03] ship without stopping

Modernise in flight

A multi-year move to SwiftUI, Combine and async/await without pausing feature delivery, and a performance programme that cut load times **40%**. Swift 6 strict concurrency, actors and structured concurrency in new code.

◆ SwiftUI migration · 40% faster loads

[04] end-to-end ownership

Credible at every layer

Native iOS (SwiftUI) and Android (Kotlin / Compose) clients on shared, versioned core libraries; NestJS / PostgreSQL / Redis backends; a Next.js console and the public web app. At Buraq I build and run **every layer myself**.

◆ 9 repos · 10k+ tests

## experience()

8 roles

iOS Team Lead @ Marn — Jahez Group

Aug 2019 → Aug 2026

RIYADH, KSA

- **Owned the architecture** of an enterprise POS used by 20k+ restaurants and cafés: an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Designed the core systems** — table management, order processing and inventory — at the heart of the product.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery.
- **Held a 99.8–99.9% crash-free rate** in production through clear module boundaries, code review and automated tests that gate merges.
- **Built CI/CD and the release process from scratch**, cutting deployment time 60%, and ran the performance programme behind 40% faster load times.
- Led and mentored a **5+ engineer iOS team** for seven years, and partnered with product, design and backend on scope and estimates.

Founder & CTO — iOS · Android · Backend · Web @ Buraq — buraqegypt.com

Mar 2026 → Present CURRENT

CAIRO · FOUNDER · BURAQ TECHNOLOGIES

Founded an Egyptian ride-hailing platform and have carried its **entire technical scope alone**: chose the stack, set the standards, and built eleven repositories from empty — a hardened backend and operations console, rider and captain apps on **iOS and Android**, a shared core package per platform, and the public web app **live at buraqegypt.com**. Every layer governed by explicit rules: idempotent mutations, double-entry accounting, acyclic module boundaries, and a CI suite that gates the merge. Now preparing a cash-only lean beta in Cairo.

› **Backend API** // NestJS · TypeScript · Mar '26 → now	Modular monolith, **22 modules** in an acyclic graph — **trip state machine**, **scored waterfall dispatch** over a Redis geo index, admin-tunable pricing across six ride tiers, **double-entry ledger** with cash and card settlement (Paymob, Fawry), scheduled and multi-stop rides, five realtime **Socket.IO gateways**, OTP / Apple / Google auth with RBAC, idempotency and audit log · **330+ endpoints** over 55 controllers, 62 entities, **2,900+ tests**, Swagger docs.
› **Ops Console** // Next.js 14 · Jul '26 → now	**29-screen** ops surface — live **MapLibre** trips map with demand heatmap, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO** · English and Arabic, 170+ tests.
› **iOS · Rider & Captain** // SwiftUI · Swift 6 · Jul '26 → now	Two native **SwiftUI** apps on a shared **BuraqCore** package (domain · networking · design system, version-pinned) — the full **trip lifecycle** on live GPS, Socket.IO feeds with fresh-token reconnect, chat, wallet and payments, **Live Activities**, Siri voice booking, on-device ID verification, captain earnings and navigation · ~240 views, **1,000+ tests**, strict concurrency.
› **Android · Captain** // Kotlin · Jetpack Compose · Aug '26 → now	Native port against **the same API contract** — 18 feature modules on convention plugins, Hilt, a pure-Kotlin domain; onboarding, the shift's foreground service and the trip end to end running against the backend · **1,500+ tests**.
› **Android · Rider** // Kotlin · Jetpack Compose · Sep '26 → now	Screen-by-screen parity port of the iOS rider app — 11 feature modules on a published **buraq-core-android** (GitHub Packages) · **2,200+ tests**, including 371 screenshot goldens in light, dark and RTL.
› **Web · buraqegypt.com** // Next.js 16 · React 19 · Oct '26 → now	Marketing site, **rider web app** with live trip tracking, **captain onboarding portal** with on-device document OCR, public trip-share pages and app links · English and Arabic (RTL), WCAG 2.2 AA, **1,800+ tests** · **live on Render since 15 Oct 2026**.
› **iOS · Buraq Schools** // SwiftUI · Swift 6 · Nov '26 → now	A second product on the same platform: **one app for parents and school-bus supervisors** — the live bus, safe handovers, an offline trip mode and an **iOS 26 Liquid Glass** interface · **1,000+ tests**; the Android port is under way.

One backend · one contract · eight client surfaces — and the company around them: incorporation, the Law 87/2018 licence path, a pricing engine fitted to 626 observed fares, the business model and investor materials, the go-live plan. 3,100+ commits, one author — so I know exactly what I will be asking of the team I hire.

Senior iOS Engineer @ HaramBlur · voluntary charity project

Dec 2025 → Jun 2026

- Designed an architecture integrating **on-device machine learning** for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

iOS Team Lead @ Ego — Ride Hailing

Mar 2018 → May 2019

RIYADH, KSA

- Developed a ride-hailing ecosystem (iOS, watchOS) serving **15M+ users** in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Senior iOS Engineer @ HungerStation — DeliveryHero

Sep 2016 → Mar 2018

KSA

- Contributed to a major food-delivery platform serving **20M+ users**.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

iOS Developer @ ALNapedh

May 2016 → Aug 2016

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

iOS & Apple Watch Developer @ Nile Design Co.

Jul 2015 → Sep 2015

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

iOS Developer @ UK Pro. Solutions Ltd.

May 2014 → May 2015

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

## projects[]

selected

[01] ai / privacy

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur, live video monitoring.

◆ AI-powered privacy browser

[02] full-stack

Buraq — Ride-Hailing

NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js · SwiftUI · Kotlin · Jetpack Compose

Egyptian ride-hailing platform I founded and built solo: a hardened NestJS backend (trip state machine, scored dispatch, double-entry ledger, five realtime gateways, 2,900+ tests), a 29-screen ops console, rider and captain apps on **iOS and Android**, and the public web app **live at buraqegypt.com**.

◆ Founder & CTO · nine repos · 10k+ tests

[03] delivery

HungerStation

UIKit · Firebase · Real-time APIs · Perf

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

[04] mobility

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · watchOS

Complete ride-hailing ecosystem: rider, driver, and Apple Watch apps. Real-time GPS and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

[05] enterprise

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 20k+ restaurants across Saudi Arabia. Offline-first architecture, complex table management, real-time orders.

★ 4.8 / 5 — 2,437 ratings

[06] community

Daleelo

MapKit · Core Location · Social SDKs

Muslim community app — prayer times, mosque finder, and community features with enhanced location + social integration.

★ 4.5 / 5 — 1,287 ratings

## stack.config

engineering_practice =
- System Design
- Automated Testing
- API Design
- Offline-first & Sync
- Realtime (WebSockets)
- Performance Profiling
- CI/CD
- Code Review
- Modular Codebases
- Versioned SDKs
- Mentoring
- Agile / Scrum

ios_2025 =
- Swift 6.0
- SwiftUI
- UIKit
- Combine
- async/await
- SwiftData
- iOS 18+

android_2025 =
- Kotlin
- Jetpack Compose
- Android
- Coroutines
- Flow
- MVVM

architecture =
- MVVM
- VIPER
- Clean Architecture
- Coordinator
- TDD

apple_ecosystem =
- watchOS
- tvOS
- macOS
- WidgetKit
- Live Activities
- App Intents

intelligence =
- Core ML
- Vision
- ARKit
- Dynamic Island
- App Tracking

backend_apis =
- NestJS
- TypeScript
- PostgreSQL
- TypeORM
- Redis
- JWT/OTP
- REST
- GraphQL
- Firebase
- CloudKit
- Socket.IO
- Swagger

tooling =
- Xcode 15+
- Fastlane
- TestFlight
- Instruments
- Git
- Claude Code
- Copilot
- Cursor

## portfolio/

FOOD & GAMING
- HungerStation
- Punters

TRANSPORTATION
- Buraq — founder
- Ego — Rider
- Ego — Driver
- Ego — Watch

ENTERPRISE
- Marn POS
- Medical Guide

SOCIAL & COMMUNITY
- Daleelo
- NU-inc
- VeeU

WATCHOS
- Ego Watch
- TaQs Weather
- TweetMe

E-COMMERCE
- iBss
- Exclusive Luxe
- Scratch

RELIGIOUS & EDU
- Afasy
- ArabNileGroup

UTILITY & TOOLS
- Unlocking Smart
- Insta
- iContacts
- First iOS app

AI & ML
- HaramBlur
- RecognizeMe
- Grad project · OpenCV

WEB
- FaceDemo
- Web Pages

## record{}

// education

B.Sc. Information Systems
Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

// certification

ISTQB Foundation Level
Egyptian Software Testing Qualification Board

// languages

Arabic	native
English	professional

## honours[]

01 Owned the architecture of a POS platform serving **20k+ restaurants** and cafés for **7 consecutive years** — through its growth from launch scale to enterprise.

02 Held a **99.8–99.9% crash-free** rate and cut deployment time **60%** with the standards, tests and CI/CD the team ran on.

03 **Founded Buraq** (buraqegypt.com), an Egyptian ride-hailing platform, and built all eleven repositories solo — backend, ops console, rider and captain apps on iOS and Android, Buraq Schools, and the public web app, live since Oct 2026 — behind **10,000+ automated tests**.

04 Built a food-delivery platform serving **20M+ users** and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) & early SwiftUI adopter (WWDC 2019).

⎇ main | ● Senior Software Engineer · iOS · Android · Backend | Cairo ⟶ Riyadh | UTF-8 | TypeScript · Swift · Kotlin | ★ 12+ yrs hands-on